

Project meeting 4: Our experiment design

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1 Deep dive: Transitional probabilities (TPs)

Imagine a “language” that consists only of the sequence

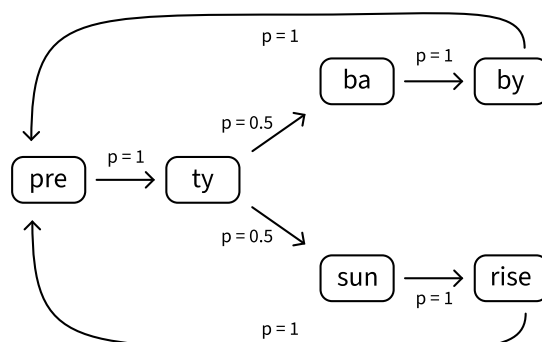
“... pretty baby pretty sunrise pretty baby pretty sunrise ...”

- After you hear “pre”, how well can you predict what syllable will come next?
- After you hear “ty”, how well can you predict what syllable will come next?
- After you hear “ba”, how well can you predict what syllable will come next?
- After you hear “by”, how well can you predict what syllable will come next?
- After you hear “sun”, how well can you predict what syllable will come next?
- After you hear “rise”, how well can you predict what syllable will come next?

To formalise these predictions, think about the probability, given the current syllable, of a particular syllable coming next.

These are conditional probabilities, written $P(\text{next syllable}|\text{current syllable})$, pronounced “the probability of the next syllable, given the current syllable”. Specifically, they are **transitional probabilities (TPs)**.

One way to showing TPs between syllables is to use a flow chart.



Another way of showing TPs is using a **transition matrix**. Transition matrices are useful when the language has lots of syllables and lots of transitions.

		to:					
		pre	ty	ba	by	sun	rise
from:	pre		1				
	ty			0.5		0.5	
	ba				1		
	by	1					
	sun						1
	rise	1					

The cells that are left empty represent a TP of 0: the transition is never observed, so we assume it's impossible in the given language.

The literature on statistical learning suggests that people segment continuous input into words by placing a word boundary where TPs are smallest (but still greater than zero).

2 Let's analyse the example experiment from our first meeting

<https://tstbl.co/764-211>

2.1 The familiarisation phase

This experiment's "language" consists of the following "words", repeated and randomly reordered: *bukadi*, *dabipu*, *paduki*, *pibaku*.

Activity all together: Fill in this language's transition matrix.

Note: The last syllable of every word can transition to the first syllable of all four words.

		to:											
		bu	ka	di	da	bi	pu	pa	du	ki	pi	ba	ku
from:	bu												
	ka												
	di												
	da												
	bi												
	pu												
	pa												
	du												
	ki												
	pi												
	ba												
	ku												

2.2 The test phase

The test phase has a structure called "two-alternative forced choice", abbreviated as 2AFC.

In each trial, participants must choose between a word that's in the language (called the "target") and a word that's not in the language (called the "foil").

target	foil
<i>bukadi</i>	<i>papida</i>
<i>dabipu</i>	<i>bikaki</i>
<i>pibaku</i>	<i>kupudi</i>

If people can distinguish targets from foils at a rate that's better than random chance (50/50), then we conclude that their learning is successful.

3 Our statistical learning experiment

I have made a prototype of our experiment on Testable. Your task is to finish implementing the experiment, using the information in this handout.

To access the prototype, send me the email address you used to register with Testable.

3.1 Our “language”

We’ll use a language that I used in the paper Pankratz et al. (2024), specifically the “suffixing” language. (A “suffix” is a part of word that attaches at the end like *-s* in *cats* or *-ed* in *walked*.)

Our language contains these eight words:

- *rukulu*
- *rukuje*
- *buzolu*
- *buzoje*
- *kodake*
- *kodavu*
- *kimuke*
- *kimuvu*

These words are “suffixing” because the “stem” (e.g., *ruku*) can appear with either one of two “suffixes” (e.g., *ruku-lu* and *ruku-je*).

- The TP between the first two syllables is 1.
- The TP between the second two syllables is 0.5.

This language structure will let us test whether differences in accuracy between TPs differs as a function of people’s autistic traits.

Activity in pairs: Fill in our language’s transition matrix.

Note: The last syllable of every word can transition to the first syllable of all eight words.

		to:											
		ru	ku	bu	zo	je	lu	ko	da	ki	mu	vu	ke
from:	ru												
	ku												
	bu												
	zo												
	je												
	lu												
	ko												
	da												
	ki												
	mu												
	vu												
	ke												

3.2 Our familiarisation phase

Participants will listen to a continuous input stream constructed out of these syllables while watching the nice space video. The familiarisation phase will last just under two minutes.

(For details about how the input stream was constructed, you can refer readers to Pankratz et al., 2024).

3.3 Our test phase

Our target items are the two syllable pairs within a word. For example, for *rukuje*, our pairs are *ruku* (TP = 1) and *kuje* (TP = 0.5).

We'll test each target twice, and we'll reuse each foil twice too. The foils contain the same syllables but in sequences that never appeared in the input stream. (In other words, TPs within the foils are all 0.)

We'll use the same audio-only 2AFC format as in the example experiment.

target	foil	TP
<i>ruku</i>	<i>rulu</i>	1.0
<i>ruku</i>	<i>vuzo</i>	1.0
<i>kuje</i>	<i>koke</i>	0.5
<i>kuje</i>	<i>muki</i>	0.5
<i>buzo</i>	<i>keda</i>	1.0
<i>buzo</i>	<i>jemu</i>	1.0
<i>zolu</i>	<i>keda</i>	0.5
<i>zolu</i>	<i>buvu</i>	0.5
<i>koda</i>	<i>luku</i>	1.0
<i>koda</i>	<i>buvu</i>	1.0
<i>davu</i>	<i>rulu</i>	0.5
<i>davu</i>	<i>koke</i>	0.5
<i>kimu</i>	<i>jemu</i>	1.0
<i>kimu</i>	<i>luku</i>	1.0
<i>muke</i>	<i>vuzo</i>	0.5
<i>muke</i>	<i>muki</i>	0.5

When implementing the experiment:

- Show participants the test trials in a random order.
- Counterbalance whether the target or foil is played first.
 - What does counterbalancing mean?
 - Since each target is tested twice, in one test trial the target should be played first, and in the other test trial the same target should be played second. (See experiment template for an illustration.)

3.4 Our autism survey

The quickest/easiest way to survey people for autistic traits is the **AQ10** (Allison et al., 2012), though note that its psychometric validity for this kind of thing is not awesome (Taylor et al., 2020).

We'll ask people to click "Agree" or "Disagree" for each of the following ten statements:

1. I often notice small sounds when others do not.
2. I usually concentrate more on the whole picture, rather than the small details.
3. I find it easy to do more than one thing at once.
4. If there is an interruption, I can switch back to what I was doing very quickly.
5. I find it easy to "read between the lines" when someone is talking to me.
6. I know how to tell if someone listening to me is getting bored.
7. When I'm reading a story, I find it difficult to work out the characters' intentions.
8. I like to collect information about categories of things (e.g., types of car, types of bird, types of train, types of plant, etc.).
9. I find it easy to work out what someone is thinking or feeling just by looking at their face.
10. I find it difficult to work out people's intentions.

Participants score a point if they answer as follows:

- "Agree" to questions 1, 7, 8, 10.
- "Disagree" to questions 2, 3, 4, 5, 6, 9.

The clinically relevant threshold is six points. Someone must score six points or higher in order to be referred for an autism diagnosis on the NHS. (But a score above six doesn't mean someone is definitely autistic—it's a necessary but not a sufficient condition.)

3.5 Other questions we'll ask at the end

- Do you have any hearing difficulties?
 - Response options: Yes, No
 - If "Yes", we'll exclude their data from the analysis
- What is your gender?
 - Response options: Woman, Man, Other
- How old are you?
 - Response options: 16–24, 25–34, 35–44, 45–54, 55–65, 65+

References

- Allison, C., Auyeung, B., & Baron-Cohen, S. (2012). Toward Brief "Red Flags" for Autism Screening: The Short Autism Spectrum Quotient and the Short Quantitative Checklist in 1,000 Cases and 3,000 Controls. *Journal of the American Academy of Child & Adolescent Psychiatry*, 51(2), 202–212.e7. <https://doi.org/10.1016/j.jaac.2011.11.003>
- Pankratz, E., Kirby, S., & Culbertson, J. (2024). Evaluating the Relative Importance of Wordhood Cues Using Statistical Learning. *Cognitive Science*, 48(3), e13429. <https://doi.org/10.1111/cogs.13429>
- Taylor, E. C., Livingston, L. A., Clutterbuck, R. A., & Shah, P. (2020). Psychometric concerns with the 10-item Autism-Spectrum Quotient (AQ10) as a measure of trait autism in the general population (C. Payne, Ed.). *Experimental Results*, 1, e3. <https://doi.org/10.1017/exp.2019.3>